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Invited article

Thermodynamic description of the Dy–Si–C system in silicon carbide ceramics

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ABSTRACT

To predict the thermal behaviors of $Dy_3Si_2C_2$ during sintering or joining process of the silicon carbide (SiC) ceramics, we established a detailed thermodynamic description of Dy–Si–C by coupling the first-principles calculations and the CALPHAD (CALculation of PHAse Diagram) approach. First-principles calculations were applied to acquire the formation enthalpies of key compounds, which can perform as end-members during optimization. We proposed a first Dy–C binary phase diagram referred to other RE-C (RE = La, Ce and Pr) systems. The Dy–Si system, previously described with the modified quasichemical model (MQM), was modeled based on the random-mixing Bragg Williams model in this work. The presently obtained thermodynamic parameters can reasonably reproduce the available experimental data and elucidate the thermal behaviors of $Dy_3Si_2C_2$. We calculated that $Dy_3Si_2C_2$ reacted with SiC via ternary eutectic reaction at 1432 °C, which was consistent with the reported experimental results. Further work of other SiC-based systems may provide foresight to explore new sintering aids in ceramics.

1. Introduction

Silicon carbide (SiC) ceramics own the merits of exceeding thermal and mechanical properties, especially the low induced radioactivity under nuclear irradiation environment [1,2]. They have vast applied prospects in high-temperature structural materials [3–5], semiconductor electronics device [6–8], nuclear reactors [9–11] etc. However, the relatively high sintering or joining temperatures have hindered the engineering applications of SiC ceramics to some extent [12].

Al–B–C [13], B_4C [14] and Al_2O_3–Y_2O_3 [15] are some frequently used additives to refine the sintering process. To elevate the efficiency and overwhelm the high temperatures during the sintering process, researchers are in an urgent need to explore suitable sintering aids. Ren et al. [16,17] prepared the core-shell structure by using a molten salt technique for the sintering of the B_4C ceramics. Inspired by the specific technique, Shao et al. [18] *in-situ* synthesized the $Y_3Si_2C_2$ coating on the surface of SiC powders to form SiC–$Y_3Si_2C_2$ core-shell structure, resulting in the relative density of 99.5% and outstanding mechanical properties. Most recently, Wan et al. [19] seamlessly joined the SiC ceramics through a sacrificial interlayer of $Dy_3Si_2C_2$ at 1500 °C.

Thermodynamic calculations via the CALPHAD (CALculation of PHAse Diagram) approach are always embedded in the development of new materials [20–22], by which, thermal behavior and densification mechanism of the $Y_3Si_2C_2$ phase have been analyzed in our previous work [23]. Likewise, thermodynamic calculations of the Dy-Si-C system with the consideration of $Dy_3Si_2C_2$ should be conducted to explore the sintering or joining process with Dy added in SiC ceramics.

In the present work, we firstly proposed the Dy–C system with respect to other RE-C (RE = La, Ce and Pr) systems [24]. Then, the random-mixing Bragg Williams model was applied to describe the Dy–Si system. First-principles calculations were applied to acquire the formation enthalpies of key compounds, which performed as end-members during optimization. As last, the Dy-Si-C system was critically optimized by considering the $Dy_3Si_2C_2$ ternary phase.

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Table 1
Crystal structure and lattice parameter data of the compounds in the Dy-Si-C system.

Phase	Space group	Phase protype	Pearson symbol	Lattice parameter (nm)	Ref.
Dy_5Si_3	*P6₃/mcm*	Mn_5Si_3	*hP16*	a = 0.8399(2); c = 0.6394(4)	[25]
Dy_5Si_4	*Pnma*	Sm_5Ge_4	*oP36*	a = 0.735(1); b = 1.445(3); c = 0.659(5)	[25]
DySi	*Cmcm*	CrB	*oS8*	a = 0.4249(3); b = 1.0505(1); c = 0.3818(1)	[25]
Dy_3Si_4	*Cmcm*	Ho_3Si_4	*oS24*	a = 0.42107(2); b = 2.3892(2); c = 0.38144(2)	[26]
α-Dy_3Si_5	*P6/mmm*	AlB_2	*hP3*	a = 0.3826(1); c = 0.4122(1)	[25]
β-Dy_3Si_5	–	–	–	–	[25]
α-$DySi_2$	*Imma*	$GdSi_2$	*oI12*	a = 0.4031(1); b = 0.3933(1); c = 1.3322(5)	[25]
β-$DySi_2$	*I4₁/amd*	$ThSi_2$	*tI12*	a = 0.403; c = 1.338	[73]
Dy_2C_3	*$I\bar{4}3d$*	Pu_2C_3	*cI40*	a = 0.8198(2) ~0.8215(1)	[41]
α-DyC_2	*I4/mmm*	CaC_2	*tI6*	a = 0.3669(2)	[41]
β-DyC_2	*Fm3m*	CaF_2	*cF12*	–	[41]
DyC_{1-x}	*Fm3m*	Defect-NaCl	-	a = 0.501	[49]
Dy_2C	*$R\bar{3}m$*	$CdCl_2$	*hR3*	a = 0.6312; α = 32.99°	[41]
Dy_3C	*Fm3m*	Fe_4N	*cF5*	a = 0.5001–0.5079 (5)	[41]
Dy_5Si_3C	*P6₃/mcm*	Hf_5Sn_3Cu	*hP18*	a = 0.838; c = 0.636	[54]
$Dy_{15}Si_9C$	*P6₃mc*	$La_{15}Ge_9Fe$	*hP50*	a = 1.4537(1); c = 0.6348(1)	[56]
$Dy_3Si_2C_2$	*Cmmm*	$Y_3Si_2C_2$	*oI36*	a = 0.3838(1); b = 1.5611(3); c = 0.4203(1)	[51]

Table 2
Summary of the experimental information of the Dy–Si system.

Type	Methods	Ref.	QM[a]
Phase diagram data	XRD, DTA, Metallography	[25]	+
Melting point ($DySi_2$)		[29–31]	–
Poly. transformation ($DySi_2$)	High-temperature XRD	[32]	–
Phase diagram data (Dy-rich)		[28]	+
Phase diagram data (Si-rich)		[27]	+
Poly. transformation ($DySi_2$)	Quantitative DTA	[34]	+
Dy_3Si_4	Single crystal X-ray	[26]	+
Enthalpy of mixing, T = 1785 K		[35]	+
C_p (α-$DySi_2$, 55–300 K)	Adiabatic calorimetry	[37]	+
Formation enthalpy, H_{1473}–H_{298}	DRC	[36]	+
H_T-H_{298} ($DySi_2$, 300–2007 K), Poly. transformation, C_p, Melting point	High temperature differential calorimetry, High temperature vacuum calorimetry, Drop calorimetry, Calculation	[33]	+
Melting point, C_p (Dy_5Si_3)	Drop calorimetry, Calculation	[38]	+
Melting point, Cp (DySi)	Drop calorimetry, Calculation	[39]	+

[a] QM: quote mode; +: used in this work; -: not used in this work.

2. Literature review

The detailed crystal structure and lattice parameter information of the compounds in the Dy-Si-C system are summarized in Table 1.

2.1. Dy–Si system

Table 2 summarizes the experimental data of the Dy–Si system on phase equilibria and thermodynamic data. Eremenko et al. [25] experimentally determined the Dy–Si phase diagram using X-ray diffraction (XRD), differential thermal analysis (DTA) and metallography methods. Seven intermetallics, viz. Dy_5Si_3, Dy_5Si_4, $DySi_{1-x}$ (x≈0.02), DySi, Dy_4Si_5, α-$DySi_{2-b}$ (b≈0.33) and α-$DySi_{2-a}$ (a≈0.18), were observed in the Dy–Si phase diagram. The Dy_5Si_3 and DySi phases melted at 1880 and 1890 °C congruently. Through the peritectic reactions at 1830, 1780, 1635 and 1430 ± 5 °C, the Dy_5Si_4, $DySi_{1-x}$, α-$DySi_{2-b}$ and α-$DySi_{2-a}$ phases formed respectively. The Dy_4Si_5 phase was decomposed at 1350 °C, which was refined to be Dy_3Si_4 by Roger et al. [26] using single crystal XRD. As for the eutectic reactions, the differences of the reaction temperatures were within 20 °C in literature [25,27,28]. Besides, there were two polymorphic transformations in the Dy–Si system. The transition temperature of α-$DySi_{2-b}$ (Dy_3Si_5) was detected to be 760 °C [25]. However, the melting and polymorphic transformation temperatures of the $DySi_2$ phase showed little resemblance in literature. Before the work of Eremenko et al. [25], the $DySi_2$ phase was reported to melt at approximate 1500 [29] and 1550 [30,31] °C, and the polymorphic transformation was determined to occur at 540 ± 40 via high-temperature XRD [32]. Gorbachuk and Bolgar [33] reported these temperatures of $DySi_2$ to be 1455 ± 40 and 464 ± 19 °C, respectively. Soon after, Gorbachuk [34] revised the polymorphic transformation temperature to be 490 °C with a quantitative DTA.

Batalin et al. [35] used high-temperature calorimetry to measure the integral enthalpy of mixing and then derived the partial enthalpy of mixing of Dy in the Dy–Si liquid at 1785 K within a limited composition range of $0<X_{Dy}<0.04$. Meschel and Kleppa [36] determined the standard formation enthalpies of the DySi, $DySi_2$ and Dy_5Si_3 phases to be −68.9 ± 2.0, −54.6 ± 2.1 and −64.0 ± 2.1 kJ/mol using direct reaction calorimetry (DRC) at 1473 ± 2 K. In addition, they reported the corresponding enthalpy increment (H_{1473}–H_{298}) as well. Gorbachuk and Bolgar [33,37] used high-temperature differential calorimetry and high-temperature vacuum calorimetry to determine the enthalpy increment (H_T-H_{298}) of $DySi_2$ in the temperature range of 300–1200 K and above 1200 K, respectively. The heat capacity of the $DySi_2$ phase at low temperatures was studied by the adiabatic method [37]. Based on low-temperature measurements, the temperature-dependent heat capacities for both α-$DySi_2$ and β-$DySi_2$ were derived [33]. In addition, Gorbachuk [38,39] proposed an equation to determine the high-temperature heat capacities for the DySi and Dy_5Si_3 phases.

The Dy–Si binary phase diagram was only optimized by Kim et al. [40] using the modified quasichemical model (MQM). Eight binary compounds were treated as stable stoichiometric phases, including two polymorphic transformations. The Dy_3Si_4 phase was calculated to decompose via eutectoid reaction at 1351 °C and the melting point of DySi was 11 °C higher than that of Dy_5Si_3. However, their model for the liquid phase is incompatible with the present work, and we have re-optimized the Dy–Si system based on the random-mixing Bragg Williams model.

2.2. Dy–C system

Up to now, no phase diagram was available for the Dy–C system. Gschneidner and Calderwood summarized the literature information of the Dy–C system before 1986 [41] and proposed the hypothetical ideal RE-C (RE = rare earth) phase diagram [42]. The space group of Dy_2C_3 was reported to be $I\bar{4}3d$, while the melting point was still unknown [43].

Table 3
Summary of the formation enthalpies (kJ/mole-atom) for the compounds in the Dy-Si-C system.

Phase	Formation enthalpy[a]	Comment[b]	Phase	Formation enthalpy[a]	Comment[b]
Dy_5Si_3	−64.0 ± 2.1 (DRC)	[36]	$DySi_2$	−54.6 ± 2.1 (DRC)	[36]
	−66.9 (FP)	TW		−50.8 (FP)	TW
	−64.4 (CAL)	TW		−54.9 (CAL)	TW
Dy_5Si_4	−72.7 (FP)	TW	Dy_3Si_4	−67.0 (FP)	TW
	−71.1 (CAL)	TW		−67.0 (CAL)	TW
DySi	−68.9 ± 2.0 (DRC)	[36]	DyC_2	−28.6 ± 2.0 (DRC)	[36]
	−80.0 (FP)	TW		−31.4 ± 10.5	[50]
	−72.0 (CAL)	TW		−15.3 ± 4.2	[72]
Dy_2C_3	−35.5 (FP)	TW		−31.0 (CAL)	TW
	−35.8 (CAL)	TW		−29.8 (FP)	TW
			$Dy_3Si_2C_2$	−61.8 (FP)	TW
				−56.4 (CAL)	TW

[a] DRC: Direct reaction calorimetry; FP: First-principles calculations; CAL: CALPHAD.
[b] TW: this work.

As for the DyC_2 phase, Krikorian et al. [44] confirmed the room temperature structure and determined the polymorphic transformation temperature to be 1295 ± 20 °C. Other investigations were also conducted and the temperatures were reported to be 1250 [45] and 1300 to 1317 [46] °C. Krikorian et al. [44] determined the DyC_2–C eutectic temperature to be 2290 ± 25 °C using the thermal analysis methods. However, Yupko et al. [47] and Kosolapova et al. [48] measured the melting point of DyC_2 to be 2250 ± 60 and 2245 °C, respectively. It was unreasonable that the melting point was lower than the eutectic temperature. Under this circumstance, Gschneidner and Calderwood [41] considered the average value 2260 ± 25 °C as the eutectic temperature. Moreover, Sommer [49] did the book review, where DyC_{1-x} and Dy_3C were provided in addition to Dy_2C_3 and DyC_2.

The standard formation enthalpy of DyC_2 was determined to be −28.6 ± 2.0 kJ/mol by Meschel and Kleppa [36] using DRC at 1473 ± 2 K. Gschneidner and Calderwood [42] reviewed the work of Anderson and Bagshaw [50] and suggested the enthalpy, entropy and gibbs energy of formation of solid DyC_2 phase at 298 K to be −31.4 kJ/mol, 34.6 J/mol·K and −41.7 kJ/mol. Up to now, no thermodynamic description for the Dy–C system can be found in literature.

Table 4
Summary of the thermodynamic parameters in the Dy-Si-C system optimized in the present work.

Phases	Thermodynamic parameters (J/mol)
Liquid: (C, Dy, Si)$_1$	${}^0L^{Liq}_{C,Dy} = -268000 + 22.2T$
	${}^1L^{Liq}_{C,Dy} = -140000$
	${}^2L^{Liq}_{C,Dy} = -24500$
	${}^0L^{Liq}_{Dy,Si} = -240162.227$
	${}^1L^{Liq}_{Dy,Si} = +6028.16945$
	${}^2L^{Liq}_{Dy,Si} = +58205.0304$
	${}^0L^{Liq}_{C,Dy,Si} = -600000$
	${}^1L^{Liq}_{C,Dy,Si} = -400000$
β-$DySi_2$: (Dy)$_1$(Si)$_2$	${}^0G^{\beta-DySi_2}_{Dy:Si} = -185093.125 + 443.340401T - 83.56TLn(T) - 0.004665T^2 + 402716T^{-1}$
α-$DySi_2$: (Dy)$_1$(Si)$_2$	${}^0G^{\alpha-DySi_2}_{Dy:Si} = -192137.525 + 397.280214T - 74.71TLn(T) - 0.009291T^2 + 450059.5T^{-1}$
Dy_5Si_3: (Dy)$_5$(Si)$_3$	${}^0G^{Dy_5Si_3}_{Dy:Si} = -578760.789 + 852.591281T - 193.96TLn(T) - 0.0256415T^2 + 214250T^{-1}$
Dy_5Si_4: (Dy)$_5$(Si)$_4$	${}^0G^{Dy_5Si_4}_{Dy:Si} = -640149.808 - 32.31T + 5G^{SER}_{Dy} + 4G^{SER}_{Si}$
β-Dy_3Si_5: (Dy)$_3$(Si)$_5$	${}^0G^{\beta-Dy_3Si_5}_{Dy:Si} = -440950.918 - 67.1700835T + 3G^{SER}_{Dy} + 5G^{SER}_{Si}$
α-Dy_3Si_5: (Dy)$_3$(Si)$_5$	${}^0G^{\alpha-Dy_3Si_5}_{Dy:Si} = -483744.473 - 25.742497T + 3G^{SER}_{Dy} + 5G^{SER}_{Si}$
DySi: (Dy)$_1$(Si)$_1$	${}^0G^{DySi}_{Dy:Si} = -157859.153 + 194.246685T - 43.72TLn(T) - 0.007405T^2 + 90137.5T^{-1}$
Dy_3Si_4: (Dy)$_3$(Si)$_4$	${}^0G^{Dy_3Si_4}_{Dy:Si} = -468642.047 - 27.7047714T + 3G^{SER}_{Dy} + 4G^{SER}_{Si}$
β-DyC_2: (Dy)$_1$(C)$_2$	${}^0G^{\beta-DyC_2}_{Dy:C} = -45000 - 27T + G^{SER}_{Dy} + 2G^{SER}_{C}$
α-DyC_2: (Dy)$_1$(C)$_2$	${}^0G^{\alpha-DyC_2}_{Dy:C} = -93000 - 3.6T + G^{SER}_{Dy} + 2G^{SER}_{C}$
Dy_2C_3: (Dy)$_2$(C)$_3$	${}^0G^{Dy_2C_3}_{Dy:C} = -179000 + 8.75T + 2G^{SER}_{Dy} + 3G^{SER}_{C}$
$Dy_3Si_2C_2$: (Dy)$_3$(Si)$_2$(C)$_2$	${}^0G^{Dy_3Si_2C_2}_{Dy:Si:C} = -432390 + 3G^{SER}_{Dy} + 2G^{SER}_{Si} + 2G^{SER}_{C}$

2.3. Dy–Si–C system

Gerdes et al. [51] reported the crystal structure of a new series of rare-earth silicide carbides with the composition $R_3Si_2C_2$ (R = Y, La–Nd, Sm, Gd–Tm). The element carbon can be interstitially alloyed into the so-called "Nowotny phase" (Mn_5Si_3-type), resulted in the formation of $Er_5Si_3C_x$, $Gd_5Si_3C_x$ and $Ho_5Si_3C_x$ [52–54]. Ritter et al. [55] reported the crystal and magnetic structure of the $R_{15}Si_9C$ compounds (R = Ho, Er, Tb). Wrubl et al. [56] reported the synthesis of the compounds $R_{15}Si_9C$ (R = Sm, Gd–Er, Y). These compounds crystallize in the $La_{15}Ge_9Fe$ structure type (see Table 1), which is an ordered superstructure of the Mn_5Si_3 type obtained from La_5Ge_3 by the insertion of one interstitial atom C per three Mn_5Si_3-type cells. Most recently, the SiC ceramics were seamlessly joined through a sacrificial interlayer of $Dy_3Si_2C_2$ at 1500 °C [19]. However, there was no precise Dy–Si–C phase diagram available for the urgent investigation.

3. Computational methods

3.1. First-principles calculations

The formation enthalpies of $Dy_3Si_2C_2$ together with the binary Dy–Si compounds were calculated based on density functional theory (DFT) [57,58] using the Vienna *ab initio* simulation package (VASP) [59–61]. The generalized gradient approximation (GGA) was used to acquire the total energy and the projected augment wave (PAW) [62,63] potentials were used to treat the balance valence electrons. In addition, the Brillouin zone ascertained after Monkhorst-Pack [64] was sampled by an appropriate k-points. A 500 eV plan-wave cut-off energy and an energy convergence criterion of 0.01 meV for the electronic structure were used. Both the volume of the unit size and atomic site were fully relaxed to find the stable state.

The enthalpy of formation for the typical $Dy_3Si_2C_2$ phase related to the composition-weighted average in its equilibrium crystal structure can be defined as:

$$\Delta H_f(Dy_3Si_2C_2) = \frac{1}{7}\left[E_{total}(Dy_3Si_2C_2) - 3E_{total}(Dy) - 2E_{total}(Si) - 2E_{total}(C)\right] \quad (1)$$

in which, $E_{total}(Dy_3Si_2C_2)$, $E_{total}(Dy)$, $E_{total}(Si)$ and $E_{total}(C)$ are the total energies of $Dy_3Si_2C_2$, Dy, Si and C.

Table 5
Calculated invariant reactions in the Dy–Si and Dy–C binary systems compared with experimental data.

Reaction	Temperature (°C)					Reaction Type
	Calc. (TW)[a]	Value	Ref.	Value	Ref.	
Dy–Si system						
L⇌DySi	1877.4	1890	[25]	1885.0	[40]	Congruent
L⇌Dy_5Si_3	1870.6	1880	[25]	1873.6	[40]	Congruent
L + Dy_5Si_3⇌Dy_5Si_4		1830	[25]			Peritectic
L + DySi⇌Dy_5Si_4	1827.7			1836.9	[40]	Peritectic
L⇌Dy_5Si_3+Dy_5Si_4	1823.6			1836.0	[40]	Eutectic
L + DySi⇌$DySi_{1-x}$		1805	[25]			Peritectic
L⇌$DySi_{1-x}$ + Dy_5Si_4		1780	[25]			
L + DySi⇌β-Dy_3Si_5	1630.3	1635	[25]	1634.7	[40]	Peritectic
L+β-Dy_3Si_5⇌β-$DySi_2$	1436.8	1430 ± 5	[25]	1436.7	[40]	Peritectic
		1550	[30,31]	1500	[29]	
		1455 ± 5	[33]			
L+(βDy)⇌(αDy)	1381.0			1381.0	[40]	Peritectic
DySi+β-Dy_3Si_5⇌Dy_3Si_4	1349.9			1350.8	[40]	Peritectoid
DySi+β-Dy_3Si_5⇌Dy_4Si_5		1350	[25]			Peritectoid
L⇌(Si)+β-$DySi_2$	1220.7	1215	[25]	1224.5	[40]	Eutectic
L⇌(αDy)+Dy_5Si_3	1165.8	1180	[25]	1186.5	[40]	Eutectic
α-Dy_3Si_5⇌β-Dy_3Si_5	759.8	760	[25]	760.0	[40]	Allotropic
α-$DySi_2$⇌β-$DySi_2$	489.9	540 ± 40	[32]	490.0	[40]	Allotropic
		464 ± 19	[33]	490	[34]	
Dy–C system						
L⇌β-DyC_2	2290.6	2250 ± 60	[47]	2245	[48]	Congruent
L⇌(C)+β-DyC_2	2270.8	2290 ± 25	[44]	2260 ± 25	[41]	Eutectic
L+β-DyC_2⇌Dy_2C_3	1621.9					Peritectic
L+(βDy)⇌(αDy)	1381.0					Peritectic
α-DyC_2⇌β-DyC_2	1295.5	1295 ± 20	[44]	1250	[45]	Allotropic
		1300–17	[46]			
L⇌(αDy)+Dy_2C_3	939.4					Eutectic

[a] TW: this work.

3.2. CALPHAD

Coupling with first-principles calculations, the CALPHAD approach was applied to thermodynamically optimize the material system. The constituent Si–C binary system was taken from Gröbner et al. [65]. The Dy–Si system, which was previously described using MQM [40], was modeled based on the random-mixing Bragg Williams model in the present calculations. As for the Dy–C system, we proposed the complete phase diagram according to other RE-C (RE = La, Ce and Pr) systems [24] due to the absence of the accurate phase diagram. With the precise descriptions for the boundary binary systems, phase equilibria and thermodynamic properties in the Dy-Si-C ternary system can be computed with the Muggianu algorithm via the standard procedure [66, 67].

4. Thermodynamic models

4.1. Unary phases

The Gibbs energy function of pure element i (i = *Dy, Si or C*) with respect to temperature $G_i^{\phi}(T) = {}^0G_i^{\phi} - H_i^{SER}$ can be expressed by the following extension:

$$ {}^0G_i^{\phi}(T) = A + BT + CT\ln T + DT^2 + ET^{-1} + FT^3 + IT^7 + JT^{-9} \quad (2)$$

where, T is the absolute temperature and the H_i^{SER} is the standard element reference (*SER*) state of the corresponding element at 25 °C and 1 bar as suggested by Scientific Group Thermodata Europe (SGTE). The pure element's Gibbs energies are extracted from Dinsdale [68].

4.2. Solution phases

The ϕ (liquid) phase in the Dy-Si-C system is described by a substitution solution model and its molar Gibbs energy is given by the formula as follows:

$$ G_m^{\phi} - H^{SER} = \sum_{i=1}^{n} x_i\,{}^0G_i^{\phi} + RT\sum_{i=1}^{n} x_i\ln(x_i) + {}^EG^{\phi} \quad (3)$$

where x_i represents the molar fraction of element i (i= *Dy, Si or C*), ${}^0G_i^{\phi}$ is the Gibbs energy for pure element i in solution phase ϕ, R is the gas constant and ${}^EG^{\phi}$ is the excess Gibbs energy of ϕ, which is described as the equation below, so-called the Redlich-Kister polynomial [69]:

$$ {}^EG^{\phi} = x_i x_j \sum_{k=0}^{n} {}^kL_{ij}^{\phi}(x_i - x_j)^k \quad (4)$$

$$ {}^kL_{ij}^{\phi} = a_k + b_k T \quad (5)$$

in which, ${}^kL_{ij}^{\phi}$ is the *kth* interaction parameter. The coefficients a_k and b_k are parameters to be optimized in this work.

4.3. Intermetallic compounds

With no convincing evidence for non-negligible homogeneity ranges, most of the binary and ternary phases in the Dy-Si-C system are modeled as stoichiometric phases. The Gibbs energies can be expressed as follow:

$$ {}^0G_m^{Dy_xSi_yC_z} - x\,{}^0H_{Dy}^{SER} - y\,{}^0H_{Si}^{SER} - z\,{}^0H_C^{SER} = A + BT + x\,{}^0G_{Dy}^{SER} + y\,{}^0G_{Si}^{SER} + z\,{}^0G_C^{SER} \quad (6)$$

where, A and B are parameters to be evaluated within work.

As for the Dy_5Si_3, DySi and $DySi_2$ phases, with the precise heat capacities descriptions, the corresponding Gibbs energy function of ϕ can be given with a fixed reference at 298.15 K by:

$$ {}^0G_i^{\phi}(T) = A + BT + CT\ln T + DT^2 + ET^{-1} \quad (7)$$

where, the related coeiffients *C, D, E* can be obtained from the heat capacity functions, A and B are parameters to be evaluated within work.

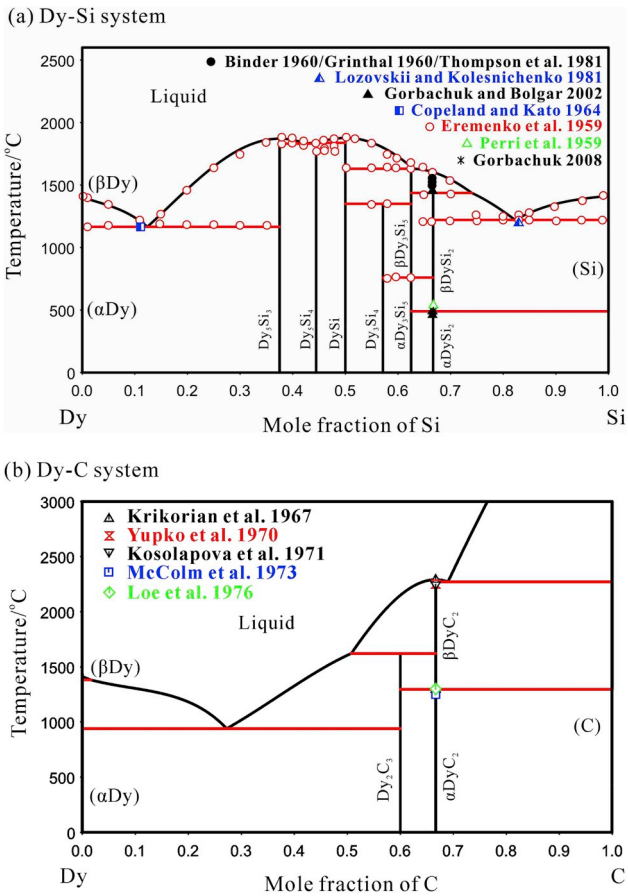

Fig. 1. Calculated binary phase diagrams in this work: (a) Dy–Si phase diagram along with experimental data [25,27–34]; (b) Dy–C phase diagram along with experimental data [44–48]. Note that the reported Dy_2C, DyC_{1-x} and Dy_3C phases [41,49] were not considered.

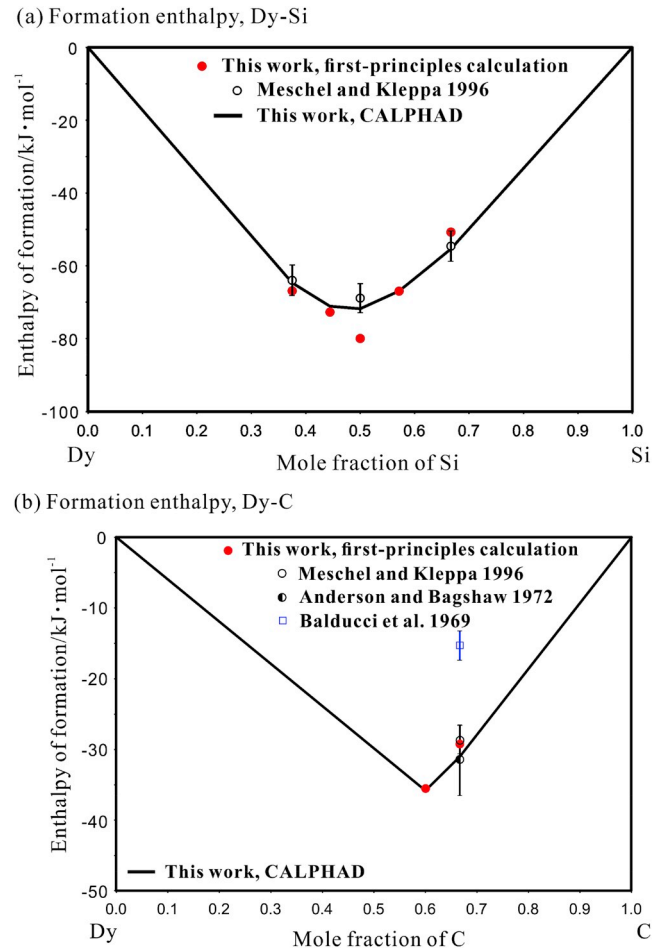

Fig. 2. The calculated standard enthalpies of formation for the (a) Dy-Si [36] and (b) Dy–C [36,50,72] binary compounds with reference states of Dy (hcp), Si (diamond) and C (Graphite).

5. Optimization procedure

The optimization procedures were conducted in the program PARROT module embedded in Thermo-Calc software [70]. The step-by-step optimization method reported by Du et al. [71] was used in this work. In the present optimization, each piece of the experimental information was critically assessed and given a certain weight (zero for those data which were not used, as listed in Table 2). The weights were adjusted during optimization until most of the used data were well agreed within the experimental uncertainty.

Abundant experimental data were available for the Dy–Si binary system, as listed in Table 2. Eremenko et al. [25] investigated the whole phase diagram using XRD, DTA and metallography methods, which was of great importance to the present optimization. Different weights were given to the melting point and polymorphic transformation temperature of the $DySi_2$ phase due to the reported wide temperature range. In addition, we used Eq. (7) to describe the Dy_5Si_3, DySi, $DySi_2$ phases due to their precise heat capacity descriptions. Formation enthalpies (see Table 3) performed as end-members during optimization.

As for the Dy–C system, no complete phase diagram was reported in the literature. The Dy_2C, DyC_{1-x} and Dy_3C phases (see Table 1) were listed in the review articles [41,49]. However, the homogeneity and stable temperature ranges are still inconclusive. Also, we were aiming at searching new sintering additives for SiC ceramics, the area of interest is the C-rich corner. Thus, we did not take these phases into consideration in the present calculations. Before optimization, we evaluated the formation enthalpy of DyC_2 [36,50,72] and calculated that of Dy_2C_3 by first-principles calculation, which we used as end-members in the first step. Then, we critically considered the melting point and the polymorphic transformation temperature of the DyC_2 phase [44–48]. At last, the phase diagram was established based on the proposed common ideal phase diagram of the RE-C systems [42].

To design the appropriate sintering process for the SiC ceramics with less experimental efforts and higher efficiency, we evaluated the thermodynamic co-relations between the SiC-$Dy_3Si_2C_2$ core-shell structure based on the phase diagram, while other ternary phases (Dy_5Si_3C, $Dy_{15}Si_9C$) [54,56] in the Dy-rich region were not considered in this work. Consequently, we thermodynamically described the $Dy_3Si_2C_2$ phase in Eq. (6) aided with first-principles calculations.

6. Results and discussion

Table 4 lists the optimized parameters for the Dy–Si, Dy–C and Dy-Si-C system together with their thermodynamic models. The calculated binary invariant reactions are summarized in Table 5.

6.1. Dy–Si binary system

Kim et al. [40] described this system using MQM, which was well accounted for the reported phase diagram data. Fig. 1a is the calculated Dy–Si binary phase diagram attached with experimental results [25, 27–34] in the present work. We also maintain a reasonable consistency with those phase diagram data.

Fig. 2a shows the calculated enthalpies of formation for the Dy–Si compounds via the CALPHAD approach compared with the

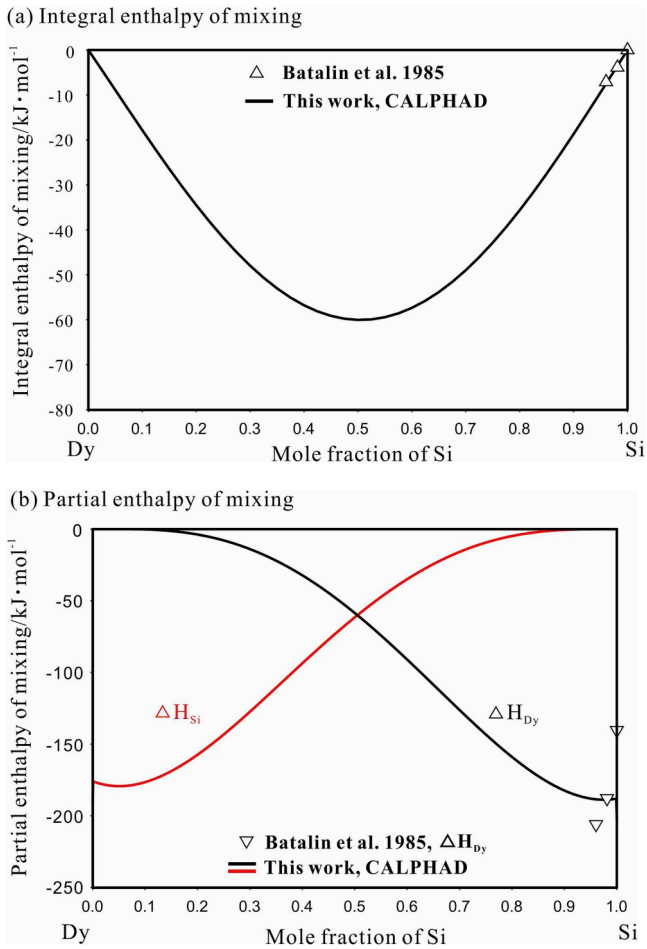

Fig. 3. Calculated enthalpy of mixing in this work: (a) Integral enthalpy and (b) partial enthalpy of mixing of Dy–Si solutions at 1785 K [35].

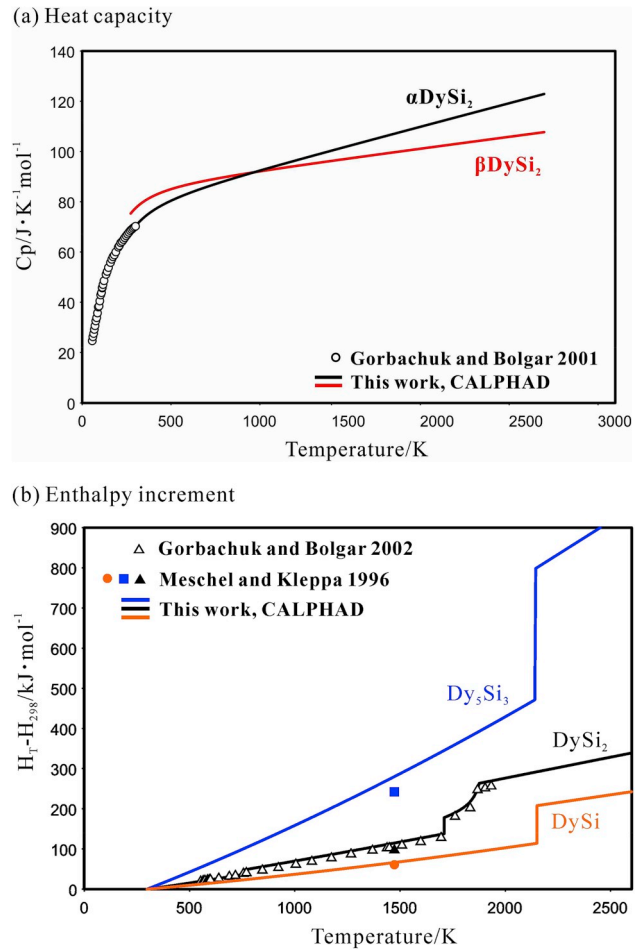

Fig. 4. Calculated thermal properties in this work: (a) Heat capacities of the α-$DySi_2$ and β-$DySi_2$ phases with experimental data [33]; (b) enthalpy increment (H_T-H_{298}) of the Dy_5Si_3, $DySi_2$ and DySi phases [33,36].

experimental results in literature [36] and the calculated data in this work (first-principles calculations). Intuitively, the experimental and calculated formation enthalpies of the $DySi_2$, Dy_3Si_5, Dy_5Si_4 and Dy_5Si_3 phases are in reasonable agreement. However, the calculated formation enthalpy for DySi is about 10 kJ/mol lower than the experimental result [36]. We have assessed to give higher weight to the experimental data in the present optimization.

Batalin et al. [35] measured the integral enthalpy of mixing and partial enthalpy of mixing of Dy in the Dy–Si liquid at 1785 K within a limited composition range of $0<X_{Dy}<0.04$. The corresponding thermodynamic data shown in Fig. 3 suit well the experiment data after adjusting the parameters of the liquid phase. Fig. 4a exhibits the calculated heat capacities of α-$DySi_2$ and β-$DySi_2$, which are derived from the work of Gorbachuk and Bolgar [37] without any modification. Fig. 4b presents the calculated enthalpy increment of Dy_5Si_3, $DySi_2$ and DySi, which agree well with those experimental data [33,36].

6.2. Dy–C binary system

Fig. 1b shows the extrapolated Dy–C binary phase diagram based on the limited experimental results [44–48] and the reported RE-C (RE = La, Ce and Pr) phase diagrams [24]. The polymorphic transformation temperature for the DyC_2 phase is calculated to be 1295.5 °C, which is reported to be 1295 ± 20 °C in literature [44]. The DyC_2-C eutectic reaction is calculated to occur at 2270.8 °C and the melting point of the high-temperature form DyC_2 phase is 2290.6 °C, which are consistent with the reviewed data [44,47,48]. Unfortunately, there is no other experimental phase diagram data for the Dy–C system. With the proposed common ideal phase diagram of the RE-C systems [42], we have optimized this system and proposed some important invariant reactions in the present work. During optimization, except for the reported melting point and the polymorphic transformation temperature of the DyC_2 phase, we have evaluated the formation enthalpy of DyC_2 and calculated that of Dy_2C_3 by first-principles calculation, which performed as important thermodynamic data in the present calculation. The Dy_2C_3 phase is predicted to form via a peritectic reaction at 1621.9 °C and there will be a eutectic reaction between Dy_2C_3 and hcp-type (Dy) at 939.4 °C.

Fig. 2b presents the calculated enthalpy of formation in the Dy–C system in this work along with literature data [36,50,72]. The calculated standard formation enthalpy of the Dy_2C_3 and DyC_2 phases are −35.8 and −31.0 kJ/mol (The entity for mol is mol of atoms) by the CALPHAD approach, respectively. As listed in Table 3, the value for Dy_2C_3 and DyC_2 are very close to the results (−35.5 and −29.8 kJ/mol) by first-principles calculations in the present work. In addition, the entropy and Gibbs energy of DyC_2 were calculated to be 39.0 J/mol·K and −48.4 kJ/mol. By considering the experimental uncertainty (10.5 kJ/mol [50] and 2.0 kJ/mol [36]) introduced from the literature, the present results are in reasonable agreement with the evaluated data (34.6 J/mol·K and −41.7 kJ/mol) reported by Gschneidner and Calderwood [42]. Therefore, an acceptable agreement can be observed by simultaneously analyzing the calculated phase diagram and thermodynamic data.

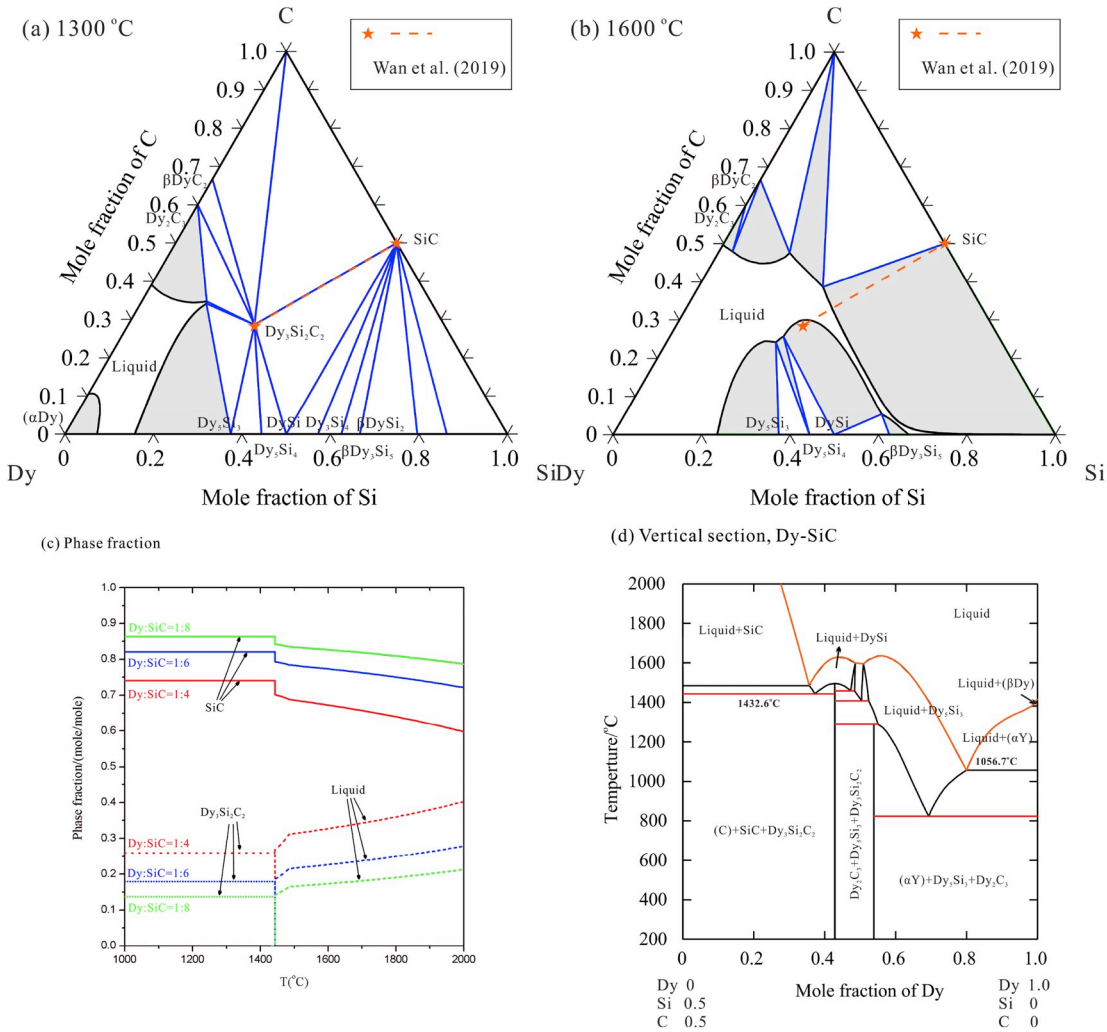

Fig. 5. Calculated phase/property diagrams of the Dy-Si-C system: (a) (b) isothermal sections at 1300 and 1600 °C, the reported seamless joining regions between $Dy_3Si_2C_2$/SiC are marked in orange dotted lines, and the star symbols represent two ends of the joining interlayers [19]; (c) phase fractions with respect to temperatures for Dy:SiC molar ratio are 1:4, 1:6 and 1:8, these ratios are expanded from the experimental samples in the Y–Si–C system [18,23]; (d) predicted vertical section along the composition of Dy:SiC = 1:1. Note that other ternary phases [54,56] were not considered in the present calculation. (For interpretation of the references to colour in this figure legend, the reader is referred to the Web version of this article.)

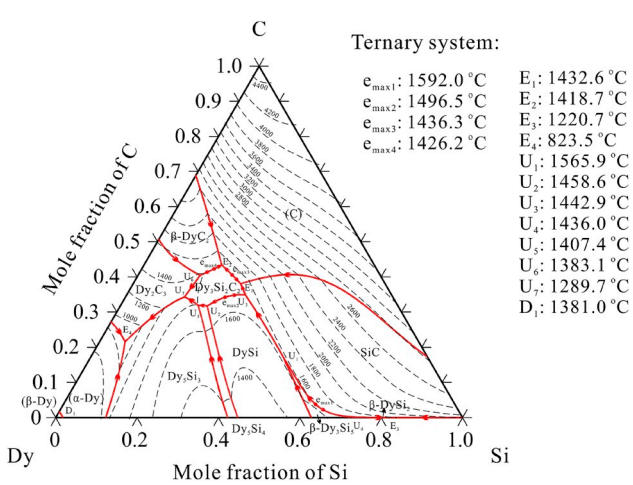

Fig. 6. Predicted liquidus projection of the Dy-Si-C system in this work.

6.3. Dy–Si–C ternary system

The formation enthalpy of the $Dy_3Si_2C_2$ phase is computed to be −61.77 kJ/mol-atom via first-principles calculations, which has performed as the end-member of $Dy_3Si_2C_2$ in the present optimization directly. It is treated as a stable stoichiometric phase in the Dy-Si-C system and the optimized parameters for the liquid and $Dy_3Si_2C_2$ phases are listed in Table 4.

Fig. 5a and b shows the calculated Dy-Si-C isothermal sections at 1300 and 1600 °C, the reported seamless joining regions between $Dy_3Si_2C_2$/SiC are marked by orange dotted lines, and the star symbols represent two ends of the joining interlayers [19]. In the SiC-rich region, the $Dy_3Si_2C_2$ phase is stable at 1300 °C and co-exists with SiC. However, the $Dy_3Si_2C_2$ phase disappears at 1600 °C, as shown in Fig. 5b. Fig. 5c presents the phase fraction diagram with the Dy:SiC ratios changed from 1:4 to 1:8, these ratios are expanded from the experimental samples in the Y–Si–C system [18,23]. At a specific temperature, the $Dy_3Si_2C_2$ phase is completely consumed, which is accompanied a notably decreased phase fraction of SiC and an increased fraction of liquid. Fig. 5d is the predicted vertical section along with the composition of Dy:SiC = 1:1, the liquidus curve is highlighted in orange. In the SiC-rich region, the liquid phase appears via the ternary eutectic reaction E_1:

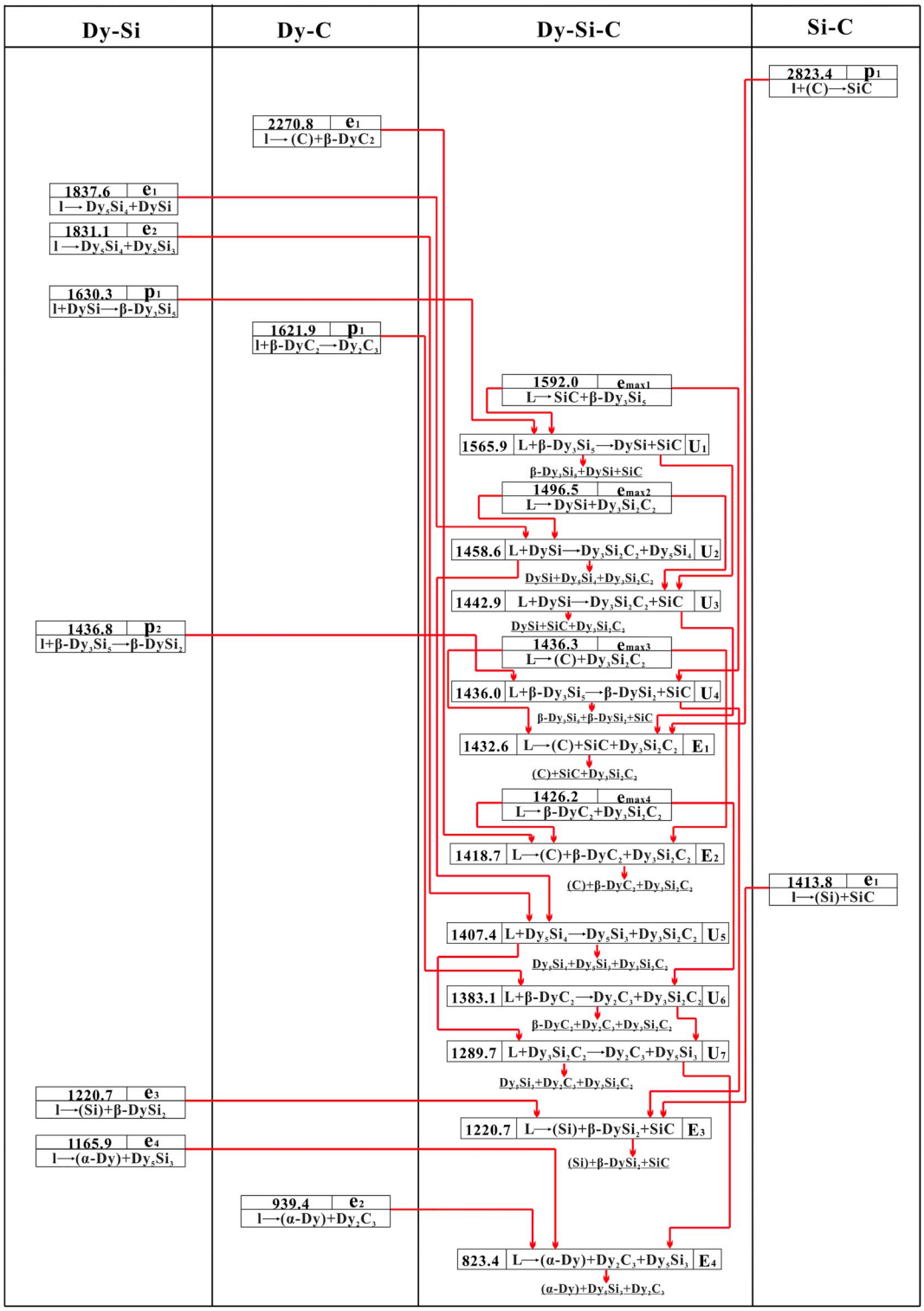

Fig. 7. Proposed liquid phase related scheil reaction scheme of the Dy-Si-C system in this work.

L⇌SiC+(C)+$Dy_3Si_2C_2$ at 1432.6 °C. The optimized thermal behaviors of the liquid and $Dy_3Si_2C_2$ phases in this work are consistent with the work of Wan et al. [19], in which, the SiC samples were joined at 1300, 1400, 1500, and 1600 °C, respectively. As for the former two samples, the residual thickness were around 3 μm, while the seamless joinings were achieved for the latter two samples. As shown in Fig. 5, with the CALPHAD approach, we know that the liquid phase started to form between the interlayer $Dy_3Si_2C_2$ and the sublayer SiC via eutectic reaction at 1432.6 °C, then, it was further squeezing out during the joining process. The fraction of the liquid phase is sufficient during the joining process because $Dy_3Si_2C_2$ reacts with SiC and liquid is formed to wet the interface. Besides, pressure is applied during the connection.

Fig. 6 is the predicted liquidus projection of the Dy-Si-C system in this work. Four ternary eutectic reactions and seven ternary transition reactions are obtained based on the present parameters, as well as four saddle points. In the Dy-rich region, the existence of the ternary eutectic

reaction E_4 (L⇌(α-Dy)+Dy_2C_3+Dy_5Si_3) makes the liquid phase to be stable low to 823.5 °C. For more details, the corresponding liquid phase related scheil reaction scheme of the Dy-Si-C system is presented in Fig. 7.

7. Conclusions

With no complete Dy–C binary phase diagram exists, the corresponding phase diagram has been proposed and calculated based on the limited experimental information and the common ideal RE-C (RE = La, Ce and Pr) phase diagrams. The Dy_2C_3 phase is predicted to form via a peritectic reaction at 1621.9 °C and there will be a eutectic reaction between Dy_2C_3 and hcp-type (Dy) at 939.4 °C. As for the Dy–Si system, we have optimized this system with the random-mixing Bragg Williams model. Formation enthalpies of the Dy–Si binary compounds computed via first-principles calculations in this work are also treated as end-members during optimization.

The Dy-Si-C system has been critically evaluated with the ternary $Dy_3Si_2C_2$ phase. A notably decreased phase fraction of SiC and an increased fraction of liquid can be estimated when the Dy:SiC ratio decreased from 1:4 to 1:8. In the SiC-rich region, the liquid phase appeared at the temperature of 1432.6 °C via the ternary eutectic reaction: L⇌SiC+(C)+$Dy_3Si_2C_2$. The presently optimized thermal behaviors of the liquid and $Dy_3Si_2C_2$ phases reveal that the high-quality sintering or joining by using $Dy_3Si_2C_2$ coatings in the SiC ceramics can be conducted above 1500 °C.

In future, the experimental Dy–C binary phase diagram can be established by preparing key alloys at specific compositions. In this case, the stabilities of DyC_{1-x} and Dy_3C in the Dy-rich region and the detailed phase relations can be evidenced. In addition, the corresponding formation enthalpies can be calculated by first-principles calculations to further confirm these results. As for the ternary system, experiments in the Dy–Si side can be conducted to study the solution behaviors of C in Dy–Si binary compounds in more details.

Data availability statement

The raw/processed data required to reproduce these findings are available to download from the corresponding author by request.

Declaration of competing interest

We declare that we do not have any commercial or associative interest that represents a conflict of interest in connection with the work submitted.

Acknowledgements

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Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.calphad.2020.101738.

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